

BFO 2020 Continuant Mereology Axioms

Continuant part of and has continuant part are inverse relations [eld-1]

$$\forall t,a,b (\text{continuantPartOf}(a,b,t) \leftrightarrow \text{hasContinuantPart}(b,a,t))$$

All independent continuants are continuant part of themselves [mcd-1]

$$\forall a,t (\text{instanceOf}(a,\text{independentContinuant},t) \rightarrow \text{continuantPartOf}(a,a,t))$$

Proper continuant part of and has proper continuant part are inverse relations [hpm-1]

$$\forall t,a,b (\text{properContinuantPartOf}(a,b,t) \leftrightarrow \text{hasProperContinuantPart}(b,a,t))$$

Exists at is disjunctive on first argument when it is a continuant [uns-1]

$$\forall p,q,r (\text{existsAt}(p,q) \wedge \text{continuantPartOf}(r,p,q) \rightarrow \text{existsAt}(r,q))$$

A fiat point has no parts other than itself [jgo-1]

$$\forall fp,t,p (\text{instanceOf}(fp,\text{fiatPoint},t) \wedge \text{continuantPartOf}(p,fp,t) \rightarrow p=fp)$$

Continuant part of is disjunctive on third argument, a temporal region [mqp-1]

$$\forall p,q,r,s (\text{continuantPartOf}(p,q,r) \wedge \text{temporalPartOf}(s,r) \rightarrow \text{continuantPartOf}(p,q,s))$$

If a has continuant part b then if a is an instance of fiat point then b is an instance of fiat point [jqd-1]

$$\forall p,q,t (\text{hasContinuantPart}(p,q,t) \wedge \text{instanceOf}(p,\text{fiatPoint},t) \rightarrow \text{instanceOf}(q,\text{fiatPoint},t))$$

If a continuant part of b then if a is an instance of spatial region then b is an instance of spatial region, and vice versa [kbr-2]

$$\begin{aligned} &\forall p,q,t (\text{continuantPartOf}(p,q,t) \\ &\rightarrow (\text{instanceOf}(p,\text{spatialRegion},t) \leftrightarrow \text{instanceOf}(q,\text{spatialRegion},t))) \end{aligned}$$

X proper continuant part of y means x is a continuant part of y but y is not continuant part of x [sls-1]

$$\begin{aligned} &\forall x,y,t (\text{properContinuantPartOf}(x,y,t) \\ &\leftrightarrow \text{continuantPartOf}(x,y,t) \wedge \neg \text{continuantPartOf}(y,x,t)) \end{aligned}$$

Proper continuant part of is disjunctive on third argument, a temporal region [vjv-1]

$$\begin{aligned} &\forall p,q,r,s (\text{properContinuantPartOf}(p,q,r) \wedge \text{temporalPartOf}(s,r) \\ &\rightarrow \text{properContinuantPartOf}(p,q,s)) \end{aligned}$$

If a continuant part of b then if a is an instance of material entity then b is an instance of material entity [dok-1]

$$\begin{aligned} &\forall p,q,t (\text{continuantPartOf}(p,q,t) \wedge \text{instanceOf}(p,\text{materialEntity},t) \\ &\rightarrow \text{instanceOf}(q,\text{materialEntity},t)) \end{aligned}$$

If a has continuant part b then if a is an instance of fiat surface then b is an instance of continuant fiat boundary [ysp-1]

$$\begin{aligned} &\forall p,q,t (\text{hasContinuantPart}(p,q,t) \wedge \text{instanceOf}(p,\text{fiatSurface},t) \\ &\rightarrow \text{instanceOf}(q,\text{continuantFiatBoundary},t)) \end{aligned}$$

If a continuant part of b then if a is an instance of independent continuant then b is an instance of independent continuant, and vice versa [cez-2]

$$\begin{aligned} &\forall p,q,t (\text{continuantPartOf}(p,q,t) \\ &\rightarrow (\text{instanceOf}(p,\text{independentContinuant},t) \leftrightarrow \text{instanceOf}(q,\text{independentContinuant},t))) \end{aligned}$$

If a has continuant part b then if a is an instance of three dimensional spatial region then b is an instance of spatial region [fzg-1]

$$\begin{aligned} &\forall p,q,t (\text{hasContinuantPart}(p,q,t) \wedge \text{instanceOf}(p,\text{threeDimensionalSpatialRegion},t) \\ &\rightarrow \text{instanceOf}(q,\text{spatialRegion},t)) \end{aligned}$$

If a has continuant part b then if a is an instance of continuant fiat boundary then b is an instance of continuant fiat boundary [ixo-1]

$$\begin{aligned} &\forall p,q,t (\text{hasContinuantPart}(p,q,t) \wedge \text{instanceOf}(p,\text{continuantFiatBoundary},t) \\ &\rightarrow \text{instanceOf}(q,\text{continuantFiatBoundary},t)) \end{aligned}$$

If a continuant part of b then if a is an instance of site then b is an instance of site or material entity [izr-1]

$$\begin{aligned} &\forall p,q,t (\text{continuantPartOf}(p,q,t) \wedge \text{instanceOf}(p,\text{site},t) \\ &\rightarrow \text{instanceOf}(q,\text{site},t) \vee \text{instanceOf}(q,\text{materialEntity},t)) \end{aligned}$$

Continuant part of is transitive at a time [plp-1]

$$\forall a,b,c,t,t2 (\text{continuantPartOf}(a,b,t) \wedge \text{continuantPartOf}(b,c,t2) \wedge \text{temporalPartOf}(t,t2) \rightarrow \text{continuantPartOf}(a,c,t))$$

Continuant part of is time indexed and has domain: continuant and range: continuant [bdd-1]

$$\forall a,b,t (\text{continuantPartOf}(a,b,t) \rightarrow \text{instanceOf}(a,\text{continuant},t) \wedge \text{instanceOf}(b,\text{continuant},t) \wedge \text{instanceOf}(t,\text{temporalRegion},t))$$

If a has continuant part b then if a is an instance of fiat line then b is an instance of fiat line or fiat point [cwp-1]

$$\forall p,q,t (\text{hasContinuantPart}(p,q,t) \wedge \text{instanceOf}(p,\text{fiatLine},t) \rightarrow \text{instanceOf}(q,\text{fiatLine},t) \vee \text{instanceOf}(q,\text{fiatPoint},t))$$

If a has continuant part b then if a is an instance of site then b is an instance of site or continuant fiat boundary [mjj-1]

$$\forall p,q,t (\text{hasContinuantPart}(p,q,t) \wedge \text{instanceOf}(p,\text{site},t) \rightarrow \text{instanceOf}(q,\text{site},t) \vee \text{instanceOf}(q,\text{continuantFiatBoundary},t))$$

Proper continuant part of is time indexed and has domain: continuant and range: continuant [kte-1]

$$\forall a,b,t (\text{properContinuantPartOf}(a,b,t) \rightarrow \text{instanceOf}(a,\text{continuant},t) \wedge \text{instanceOf}(b,\text{continuant},t) \wedge \text{instanceOf}(t,\text{temporalRegion},t))$$

If a has continuant part b then if a is an instance of zero dimensional spatial region then b is an instance of zero dimensional spatial region [bfv-1]

$$\forall p,q,t (\text{hasContinuantPart}(p,q,t) \wedge \text{instanceOf}(p,\text{zeroDimensionalSpatialRegion},t) \rightarrow \text{instanceOf}(q,\text{zeroDimensionalSpatialRegion},t))$$

Proper continuant part of is transitive at a time [xpg-1]

$$\forall a,b,c,t,t2 (\text{properContinuantPartOf}(a,b,t) \wedge \text{properContinuantPartOf}(b,c,t2) \wedge \text{temporalPartOf}(t,t2) \rightarrow \text{properContinuantPartOf}(a,c,t))$$

A fiat line occupies a one dimensional spatial region [kcq-2]

$$\forall x,t (\text{instanceOf}(x,\text{fiatLine},t) \rightarrow \exists s,tp (\text{temporalPartOf}(tp,t) \wedge \text{occupiesSpatialRegion}(x,s,tp) \wedge \text{instanceOf}(s,\text{oneDimensionalSpatialRegion},tp)))$$

A fiat point occupies a zero dimensional spatial region [alm-2]

$$\forall x,t (\text{instanceOf}(x,\text{fiatPoint},t) \rightarrow \exists tp,s (\text{temporalPartOf}(tp,t) \wedge \text{occupiesSpatialRegion}(x,s,tp) \wedge \text{instanceOf}(s,\text{zeroDimensionalSpatialRegion},tp)))$$

A fiat surface occupies a two dimensional spatial region [fpl-2]

$$\forall x,t (\text{instanceOf}(x,\text{fiatSurface},t) \rightarrow \exists s,tp (\text{temporalPartOf}(tp,t) \wedge \text{occupiesSpatialRegion}(x,s,tp) \wedge \text{instanceOf}(s,\text{twoDimensionalSpatialRegion},tp)))$$

If a has continuant part b then if a is an instance of material entity then b is an instance of site or continuant fiat boundary or material entity [mic-1]

$$\forall p,q,t (\text{hasContinuantPart}(p,q,t) \wedge \text{instanceOf}(p,\text{materialEntity},t) \rightarrow \text{instanceOf}(q,\text{site},t) \vee \text{instanceOf}(q,\text{continuantFiatBoundary},t) \vee \text{instanceOf}(q,\text{materialEntity},t))$$

If a has continuant part b then if a is an instance of one dimensional spatial region then b is an instance of one dimensional spatial region or zero dimensional spatial region [wne-1]

$$\forall p,q,t (\text{hasContinuantPart}(p,q,t) \wedge \text{instanceOf}(p,\text{oneDimensionalSpatialRegion},t) \rightarrow \text{instanceOf}(q,\text{oneDimensionalSpatialRegion},t) \vee \text{instanceOf}(q,\text{zeroDimensionalSpatialRegion},t))$$

The dimensionality of the region of something occupying a one dimensional spatial region does not change [qfe-1]

$$\forall m,s (\exists t (\text{occupiesSpatialRegion}(m,s,t) \wedge \text{instanceOf}(s,\text{oneDimensionalSpatialRegion},t)) \rightarrow \forall t1,s1 (\text{occupiesSpatialRegion}(m,s1,t1) \rightarrow \text{instanceOf}(s1,\text{oneDimensionalSpatialRegion},t1)))$$

The dimensionality of the region of something occupying a two dimensional spatial region does not change [dor-1]

$$\forall m,s (\exists t (\text{occupiesSpatialRegion}(m,s,t) \wedge \text{instanceOf}(s,\text{twoDimensionalSpatialRegion},t)) \rightarrow \forall t1,s1 (\text{occupiesSpatialRegion}(m,s1,t1) \rightarrow \text{instanceOf}(s1,\text{twoDimensionalSpatialRegion},t1)))$$

The dimensionality of the region of something occupying a zero dimensional spatial region does not change [fok-1]

$$\begin{aligned} &\forall m,s (\exists t (\text{occupiesSpatialRegion}(m,s,t) \wedge \text{instanceOf}(s,\text{zeroDimensionalSpatialRegion},t)) \\ &\quad \rightarrow \forall t1,s1 (\text{occupiesSpatialRegion}(m,s1,t1) \\ &\quad \rightarrow \text{instanceOf}(s1,\text{zeroDimensionalSpatialRegion},t1))) \end{aligned}$$

The dimensionality of the region of something occupying a three dimensional spatial region does not change [rlf-1]

$$\begin{aligned} &\forall m,s (\exists t (\text{occupiesSpatialRegion}(m,s,t) \wedge \text{instanceOf}(s,\text{threeDimensionalSpatialRegion},t)) \\ &\quad \rightarrow \forall t1,s1 (\text{occupiesSpatialRegion}(m,s1,t1) \\ &\quad \rightarrow \text{instanceOf}(s1,\text{threeDimensionalSpatialRegion},t1))) \end{aligned}$$

If a material entity has a proper part, then at least one of its proper parts is not an immaterial entity [adm-1]

$$\begin{aligned} &\forall m,t (\text{instanceOf}(m,\text{materialEntity},t) \wedge (\exists mp (\text{continuantPartOf}(mp,m,t) \wedge mp \neq m)) \\ &\quad \rightarrow \exists mp (mp \neq m \wedge \text{continuantPartOf}(mp,m,t) \wedge \neg \text{instanceOf}(mp,\text{immaterialEntity},t))) \end{aligned}$$

If a has continuant part b then if a is an instance of two dimensional spatial region then b is an instance of two dimensional spatial region or one dimensional spatial region or zero dimensional spatial region [hbn-1]

$$\begin{aligned} &\forall p,q,t (\text{hasContinuantPart}(p,q,t) \wedge \text{instanceOf}(p,\text{twoDimensionalSpatialRegion},t) \\ &\quad \rightarrow \text{instanceOf}(q,\text{twoDimensionalSpatialRegion},t) \\ &\quad \vee \text{instanceOf}(q,\text{oneDimensionalSpatialRegion},t) \\ &\quad \vee \text{instanceOf}(q,\text{zeroDimensionalSpatialRegion},t)) \end{aligned}$$

If at any time that two non object aggregates exist each is part of the other, then they are identical [tab-1]

$$\begin{aligned} &\forall a,b (\exists t (\text{instanceOf}(a,\text{independentContinuant},t) \wedge \neg \text{instanceOf}(a,\text{objectAggregate},t) \\ &\quad \wedge \text{instanceOf}(b,\text{independentContinuant},t) \wedge \neg \text{instanceOf}(b,\text{objectAggregate},t) \\ &\quad \wedge \text{continuantPartOf}(a,b,t) \wedge \text{continuantPartOf}(b,a,t)) \\ &\quad \rightarrow a=b) \end{aligned}$$

If at all times that two object aggregates exist each is part of the other, then they are identical [glc-3]

$$\begin{aligned} &\forall a,b ((\exists t (\text{instanceOf}(a,\text{objectAggregate},t) \wedge \text{continuantPartOf}(a,b,t) \\ &\quad \wedge \text{continuantPartOf}(b,a,t))) \\ &\quad \wedge (\forall t (\text{existsAt}(a,t) \leftrightarrow \text{existsAt}(b,t))) \\ &\quad \wedge (\forall t (\text{existsAt}(a,t) \\ &\quad \rightarrow \text{instanceOf}(a,\text{objectAggregate},t) \wedge \text{instanceOf}(b,\text{objectAggregate},t) \\ &\quad \wedge \text{continuantPartOf}(a,b,t) \wedge \text{continuantPartOf}(b,a,t))) \\ &\quad \rightarrow a=b) \end{aligned}$$

The continuant parthood relation satisfies weak supplementation [fyf-2]

$$\begin{aligned} &\forall t,x,y (\text{instanceOf}(x,\text{continuant},t) \wedge \text{instanceOf}(y,\text{continuant},t) \\ &\quad \wedge \text{instanceOf}(t,\text{temporalRegion},t) \\ &\quad \rightarrow (\text{properContinuantPartOf}(x,y,t) \\ &\quad \rightarrow \exists z (\text{instanceOf}(z,\text{continuant},t) \wedge \text{properContinuantPartOf}(z,y,t) \\ &\quad \quad \wedge \neg (\exists \text{overlap} (\text{instanceOf}(\text{overlap},\text{continuant},t) \\ &\quad \quad \quad \wedge \text{continuantPartOf}(\text{overlap},x,t) \\ &\quad \quad \quad \wedge \text{continuantPartOf}(\text{overlap},z,t)))))) \end{aligned}$$

Continuant part of has a unique product at a time [gzc-1]

$$\begin{aligned} &\forall x,y,t (\text{instanceOf}(x,\text{continuant},t) \wedge \text{instanceOf}(y,\text{continuant},t) \\ &\quad \wedge \text{instanceOf}(t,\text{temporalRegion},t) \\ &\quad \rightarrow (\exists \text{overlap} (\text{instanceOf}(\text{overlap},\text{continuant},t) \wedge \text{continuantPartOf}(\text{overlap},x,t) \\ &\quad \quad \wedge \text{continuantPartOf}(\text{overlap},y,t)) \\ &\quad \rightarrow \exists \text{overlap} (\text{instanceOf}(\text{overlap},\text{continuant},t) \\ &\quad \quad \wedge (\forall w (\text{instanceOf}(w,\text{continuant},t) \\ &\quad \quad \rightarrow (\text{continuantPartOf}(w,\text{overlap},t) \\ &\quad \quad \leftrightarrow \text{continuantPartOf}(w,x,t) \wedge \text{continuantPartOf}(w,y,t)))))) \end{aligned}$$
